

B.Tech.

Semester - III

Branch: CE, Chem, EE, ECE, Mech & MME

Subject: Mathematics-III

Code: MTH-231

Max. Marks: 20

Time: 90 Minutes

Note:

- All questions are compulsory and carry equal marks. Usual notations are used.
- Assume missing data if any.

1. Find the value of x in $(0, 2)$, correct to four decimal places such that $e^{x^6} = 5$.
2. Find the solution of the given system of linear equations using Gaussian elimination.

$$x + 2y + 3z - u = 10$$

$$2x + 3y - 3z - u = 1$$

$$2x - y + 2z + 3u = 7$$

$$3x + 2y - 4z + 3u = 2$$

3. Solve the given system of linear equations correct to four decimal places using the Gauss-Seidel method.

$$3x + 9y - 2z = 11$$

$$4x + 2y + 13z = 24$$

$$4x - 2y + z = -8$$

4. For a hot air balloon the distance (in nautical miles) of the visible horizon is given below for different heights (in feet) above sea level. Find the value of y when the balloon is at a height of 410 ft.

height, x	100	150	200	250	300	350	400
distance, y	10.63	13.03	15.04	16.81	18.42	19.90	21.27

5. The following table gives the values of $y = e^{1-x}$ for certain equidistant values of x . Find the interpolating polynomial and then the value of $e^{0.644}$ using Bessel's interpolating formula, correct to six decimal places.

x	0.32	0.33	0.34	0.35	0.36	0.37	0.38	0.39
y	1.973878	1.954237	1.934792	1.915541	1.896481	1.877610	1.858928	1.840431